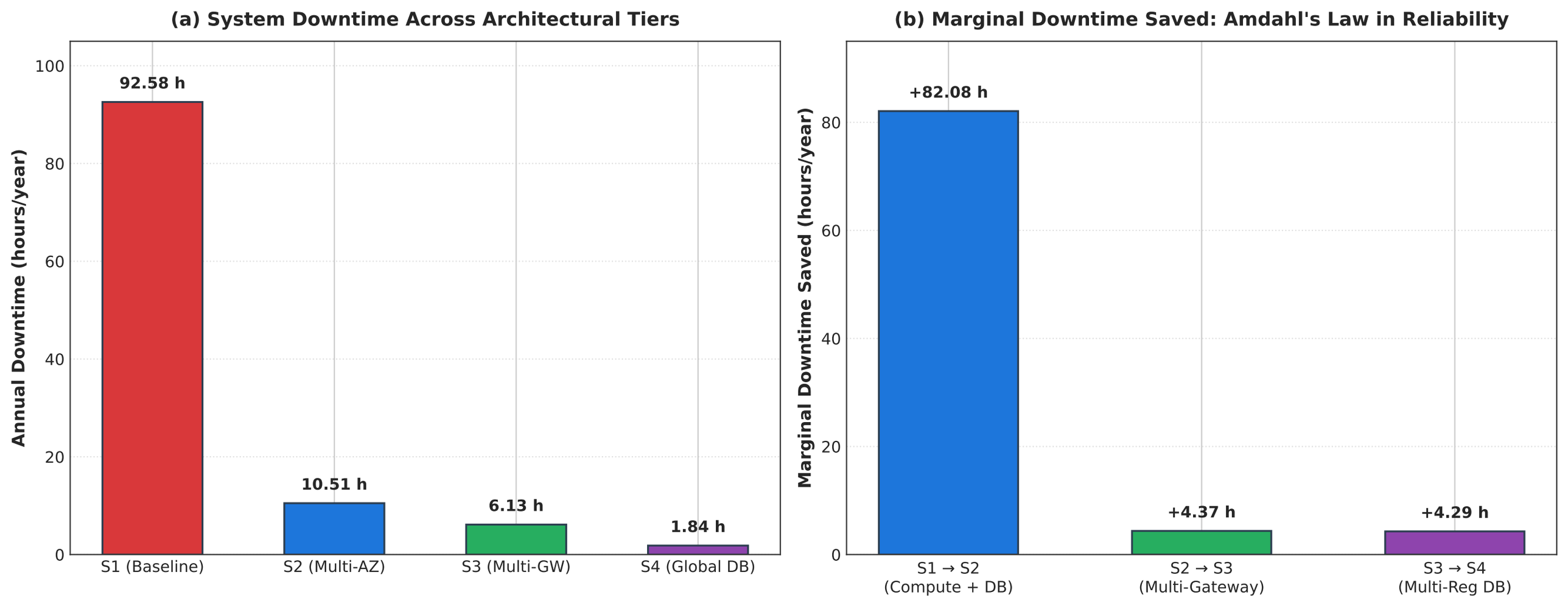
(a) System Downtime Across Architectural Tiers
Annual Downtime (hours/year)
0
20
40
60
80
100
92.58 h
10.51 h
6.13 h
1.84 h
S1 (Baseline)
S2 (Multi-AZ)
S3 (Multi-GW)
S4 (Global DB)
(b) Marginal Downtime Saved: Amdahl's Law in Reliability
Marginal Downtime Saved (hours/year)
0
20
40
60
80
+82.08 h
+4.37 h
+4.29 h
S1 → S2
(Compute + DB)
S2 → S3
(Multi-Gateway)
S3 → S4
(Multi-Reg DB)